Machine Learning Frameworks for College Football Play Classification

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***Abstract*—With the rise of data analytics in football, college football data analysis has become a critical part of a team's preparation to understand an opposing team's tendencies, coaching strategies, situational play calling, and more. Current data analysis for college football focuses on the visual presentation of statistics, and tendencies of opposing teams, instead of predictive modeling of future plays. In this paper, we propose a study on how different machine learning algorithms can be leveraged to predict a team's next play given a team's data, and game situation. This model will use data from all power 4 teams, those in the ACC, Big Ten, Big 12, and SEC. This data set is custom created, using two different scrapers to get data from ESPN's APIs. The models will classify plays as either a run or a pass before the play is called, based off a whole host of data including, down, distance, score, score differential, quarter, time, previous sequence of plays, and a running total of team statistics for the season. Multiple different machine learning algorithms including Random Forest, K-Nearest Neighbor, Naive Bayes, Logistic Regression, Gradient Boosting, and Conditional Inference Trees are trained and evaluated to determine the most effective model for predicting play outcomes. This framework provides a data-driven predictive model for college football plays.**

***Index Terms*—Play prediction, machine learning, Random Forest, Gradient Boosting, K-Nearest Neighbor, Naive Bayes, Logistic Regression, Conditional Inference Trees, predictive analytics, sports data analytics.**

I. Introduction

The integration of data analytics in sports has been on the rise though the past few decades, leading to higher optimization, particularly in the National Football League, while FBS College Football has adopted many of the same strategies. The growth of these data analytics has transformed how teams prepare, strategize, and compete. This is usually shown in the form of data visualization, play call decision, and tendencies. This, however, lacks the application of any predictive analytics, despite an overwhelming amount of data that can be used such as down, distance, yardage, score, quarter, time, and a running total of team statistics. This gap in analytics allows machine learning methods to be used to predict upcoming plays, helping offer a deeper understanding of team strategies and tendencies.

The objective of this paper is to develop data-driven frameworks to predict whether an offensive team's next play will be a run or a pass. This is done through data collected from the 2024 college football season. The model will use and develop patterns based off down, distance, quarter, score differential, field position, and previous plays. By capturing game conditions and context of previous plays, this framework emulates how coaches adjust play calling across different situations throughout games. While some prior works have modeled play calling as a sequential decision process using hidden Markov models to capture drive-level temporal patterns [5], our approach instead relies solely on situational pre-snap variables, enabling accurate prediction without requiring play-history context. This paper will help advance football analytics to include predictive machine learning for college football.

Prior research in machine learning classification has shown the effectiveness of a variety of algorithms in predicting outcomes across different domains. Studies in education, healthcare, and environmental modeling have successfully applied Random Forest, Gradient Boosting, Logistic Regression, K-Nearest Neighbors, and Naive Bayes to classify complex datasets. These works consistently demonstrate the importance of evaluating models using metrics such as accuracy, precision, recall, F1-score, and situational or slice-based analyses to understand performance under varying conditions. This body of research highlights the potential for applying similar classification frameworks to college football, where capturing context and situational factors can improve predictions of play types.

To address these limitations, this paper proposes machine learning frameworks that collect, process, and model play data into a binary prediction on the next play. Two Python scrapers are used to gather data for the models. First is a game IDs scraper for ESPN's website and a second is to gather the play data from ESPN's API. This dataset forms the basis of training, giving the models context to down, distance, score differential, quarter, time, previous sequence of plays, and a running total of team statistics for the season. With this context, the models can predict the outcome of the upcoming play. Models such as Random Forest, K-Nearest Neighbor, Naive Bayes, Logistic Regression, Gradient Boosting, and Conditional Inference Trees are used to predict plays. These models are then evaluated based on overall and situational accuracy. This will involve situational evaluation by the down, by the quarter, two-minute situations, and score context helping display where each model thrives or falls short. Specifically, the models are evaluated using a combination of binary cross-entropy loss, Brier score, and calibration plots.

The contributions of this paper include data collection, creating machine learning frameworks for play classification, and evaluation of the models created. First, it is creating an easy to use, and widely accessible data collection method for others to use. This is done through ESPN's APIs which are free and publicly available. This allows future work to use our data or create their own data collection by tweaking our scrapers. This data is automatically created into an easy-to-use JSON file that can be used for models to train off. This also creates multiple different models that can be used to predict future plays. These models are critical for this piece, allowing us to evaluate each model on various metrics including game situations and overall accuracy. With a wealth of data, why not try to predict the next play? This trend reflects a broader movement in football research, where machine learning is increasingly adopted to model game dynamics, optimize coaching decisions, and enhance predictive capabilities [6].

II. Related Work

A recent survey highlights the growing adoption of machine learning in football, demonstrating its use across a wide range of applications including player performance evaluation, automated tracking and indexing, pre-snap formation recognition, and strategic decision-making for play calling [6]. This breadth of application underscores the potential for predictive analytics to transform both professional and college football by enabling data-driven insights into team tendencies and game strategy. Teich, Lutz, and Kassarnig developed a machine learning model to predict the outcomes of individual plays using pre-snap context for the NFL [3]. Their focus was predicting the play's success which they quantified through the result of a given play. In their research they trained and tracked the success of several algorithms – decision trees, k-nearest neighbor, linear regression, support vector regression linear and radial basis, and feedforward neural networks. The best performing approach was support vector regression with a radial basis function kernel; this was determined by producing the lowest mean error and highest predictive stability across the team's validation sets.

Nadlin developed a model for predicting offensive play types using 14 descriptive features [1]. From these features he creates the following targets: binary (run or pass), 3-class (run, short pass, deep pass) and 4-class (outside run, middle run, short pass, deep pass). He used several learning algorithms to accomplish this: logistic regression, support vector machines, and multilayer perceptrons. The models were tested using an 80/20 split and the best performing algorithm was logistic regression which had a precision of 83% on models trained separately by down. Newman demonstrated that deep learning can be applied to automatically analyze pre-play formations, enabling automated feature extraction from complex visual and spatial data [8]. This approach highlights the potential of machine learning to capture nuanced pre-snap context that can improve play-call prediction beyond traditional statistical features. This accuracy score was achieved by dividing true positive results by the sum of true and false positives.

In Herrmann, Schmidt, and Smith's study the performance of various classification algorithms, some of which have been mentioned in the prior related works, are evaluated in predicting educational task outcomes [4]. The nuances of each algorithm are also explored in depth. The authors used a variety of evaluation metrics including accuracy, recall, F1-score and AUC to measure the effectiveness of their models. However, the most important aspect of their research is how they highlight how different algorithms perform under distinct conditions.

III. Approach

The proposed framework consists of four key parts: data gathering, preprocessing, machine learning development, and evaluation. This will result in automated data collection with custom data scrapers, structuring data from the scrapers, applying machine learning techniques to the data, and evaluating the data. Fig. 1 illustrates the complete system architecture.

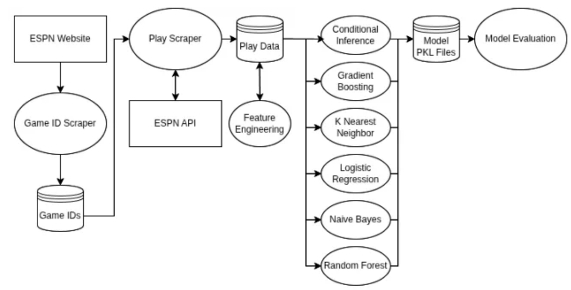

Fig. 1. System architecture showing the data collection pipeline from ESPN through feature engineering to model training and evaluation.

For data collection, two data scrapers are used. The first is a game ID scraper; this automatically scrapes all game IDs from the URL of all power 4 games. This will create a list of all the games the teams will play, and it will ignore any duplicate game IDs. This data is then utilized in the play data scraper. This will take in a list of all the game IDs that previously produced and plug them into the URL for ESPN's API. This is then used to gather information from the API. The resulting dataset is then stored in large JSON files, with thousands of plays to feed into the machine learning algorithms.

The models we plan to implement will include Random Forest, K-Nearest Neighbor, Naive Bayes, Logistic Regression, Gradient Boosting, and Conditional Inference Trees. Each of these models, like Sun's approach, was picked specifically for their own unique advantages in understanding the patterns within our dataset and identifying the best-performing approach for predicting play types [6]. This then leads into model development, which will include several different machine learning algorithms for play type classification. Each model will be trained on an 80-20 split for training and testing.

To evaluate these models, situational and overall accuracy must be accounted for. We will also use precision, recall, F1-scores, and confusion matrices to assess classification performance. This will involve situational slices by the down, by the quarter, two-minute situations, and score context. This will give an overall view of where each model is doing well or doing poorly. Further, a loss function can be implemented. Binary cross-entropy loss and Brier score help quantify how well each model is doing. This can then be expanded into comparisons between models. This will give an overall view of which models excel at several aspects of prediction.

IV. Experiments and Results

To compare each model fairly, each algorithm was trained on the same 2024 plays dataset, that was created using the scrapers mentioned earlier in the paper. This dataset was then used to calculate standard performance metrics, including accuracy, precision, recall, and F1-score, which are commonly used for evaluating classification algorithms [4]. Each model was further evaluated across situational contexts, including down, distance, field territory, game period, score differential, red zone, and goal line. Such sequence-based and situational analyses have been shown to capture nuances in play calling and improve the understanding of model strengths and weaknesses [5]. Each model was then also compared using prediction accuracy using confusion matrices where the darker shades of blue indicate higher accuracy compared to lighter shades. For the highest accuracy, the top left and bottom right should be as dark as possible as they represent true pass, and true run.

A. Accuracy Comparison

For accuracy comparison, accuracy is calculated by the equation:

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{Actual Results}} \tag{1}$$

Within this analysis, it is found that Random Forest, Conditional Inference Forest, and Gradient Boosting perform the best, with relatively similar results ranging from 0.635–0.643. The k-nearest neighbor lags a bit behind, with logistic regression and naive bayes lagging the rest with naive bayes performing worse at 0.588. Overall, each model is decently similar in range, which will be seen in all other overall performance charts. Fig. 2 shows the accuracy comparison across all models.

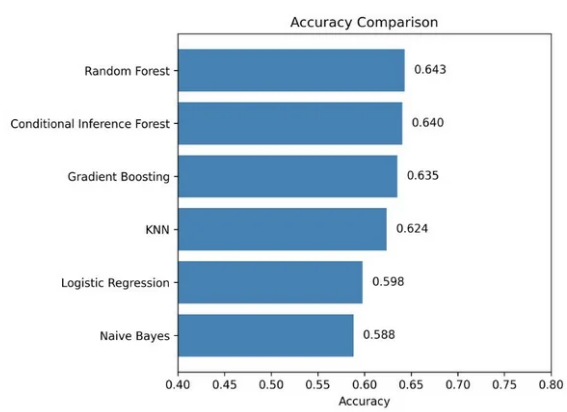

Fig. 2. Accuracy comparison across all six machine learning models.

B. Precision Comparison

For precision comparison, precision is calculated by the equation:

$$\text{Precision} = \frac{\text{True Positive}}{\text{Actual Results}} \tag{2}$$

As shown in our analysis, the Random Forest, Conditional Inference Forest, and Gradient Boosting models again have the highest precision values, ranging from 0.636 to 0.646. K-nearest neighbor performs slightly below this group, while Logistic Regression and Naïve Bayes achieve noticeably lower precision, with Naïve Bayes ranking last at 0.592. Like accuracy, the precision values remain tightly grouped, indicating that each model makes a comparable proportion of correct positive predictions. Fig. 3 shows the precision comparison across all models.

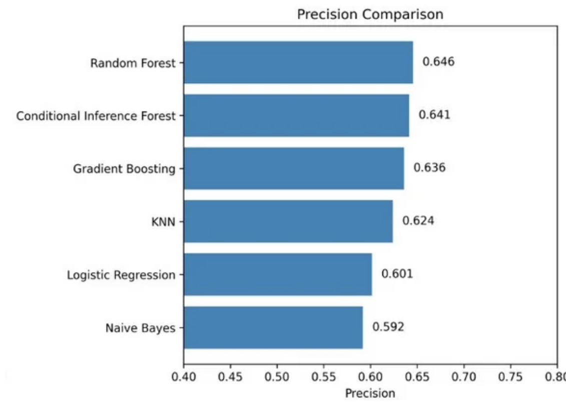

Fig. 3. Precision comparison across all six machine learning models.

C. Recall Comparison

For recall comparison, recall is calculated by the equation:

$$\text{Recall} = \frac{\text{True Positive}}{\text{Predicted Result}} \tag{3}$$

The Random Forest, Conditional Inference Forest, and Gradient Boosting models achieve the highest recall values, ranging from 0.635 to 0.643. The k-nearest neighbor performs slightly lower at 0.624, while Logistic Regression and Naïve Bayes again show the weakest performance, with Naïve Bayes obtaining the lowest recall at 0.588. Like the accuracy and precision metrics, the recall values remain closely grouped due to the balanced nature of the dataset and the symmetric misclassification patterns across the model. Fig. 4 shows the recall comparison across all models.

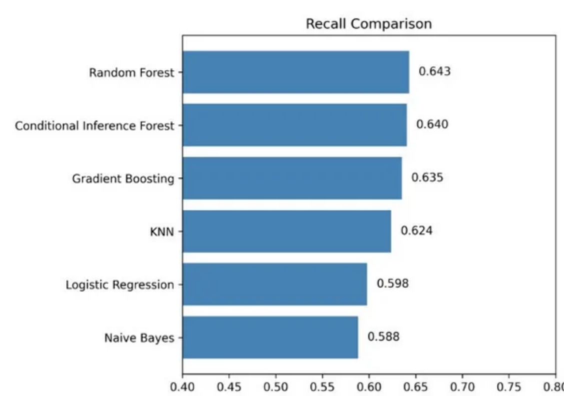

Fig. 4. Recall comparison across all six machine learning models.

D. F1-Score Comparison

For F1-score comparison, F1-score is calculated as the harmonic mean of precision and recall. The Random Forest, Conditional Inference Forest, and Gradient Boosting models obtain the highest F1-scores, ranging from 0.635 to 0.642. The k-nearest neighbor performs slightly lower at 0.624, while Logistic Regression and Naïve Bayes again yield the weakest overall performance, with Naïve Bayes achieving the lowest F1-score of 0.584. Similar to the other metrics, the F1-scores across all models remain closely grouped due to the balanced dataset and consistent error distribution between run and pass predictions. Fig. 5 shows the F1-score comparison across all models.

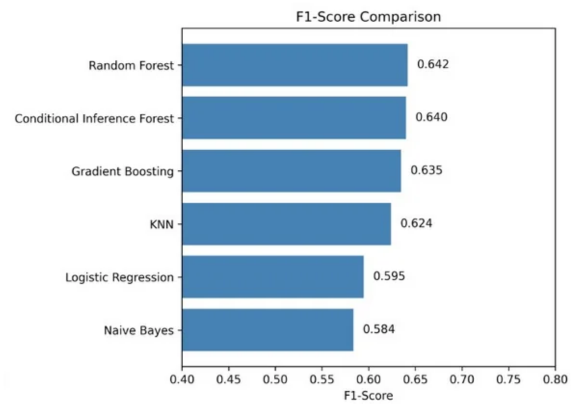

Fig. 5. F1-score comparison across all six machine learning models.

Overall, each of the metrics we have gone over so far have shown a clear hierarchy in overall performance between the models with that being Random Forest, Conditional Inference Forest, Gradient Boosting, K-Nearest Neighbor, Logistic Regression, and Naïve Bayes. This will be expanded in the analysis section.

E. Confusion Matrix Analysis

A confusion matrix provides a detailed breakdown of how each model's classification performance by comparing what the model believed the play to be to what the actual play was. This uses a y-axis which represents the true label for the play, and an x-axis of the prediction. This is then broken down to run and pass for each axis. This results in the top left corner representing a true pass, where the play was a pass, and the prediction was a pass. The top right represents a scenario where the play was a pass, and the model predicted a run. The bottom left represents a run play where the model predicted a pass. Lastly, the bottom right represents a run play where the play was a run. The color intensity corresponds to the frequency of predictions, with darker shades indicating a higher frequency. By using these figures, the distribution of values across each matrix gives deeper insights into the specific strengths and weaknesses of each classifier that are not evident from accuracy, precision, or recall alone. Fig. 6 shows the confusion matrices for all six models.

K-Nearest Neighbor shows a mid-level of performance, where it accurately gets 3,823 pass plays, and 3927 run plays correctly. It predicts 2,402 pass plays incorrect and 2,275 runs

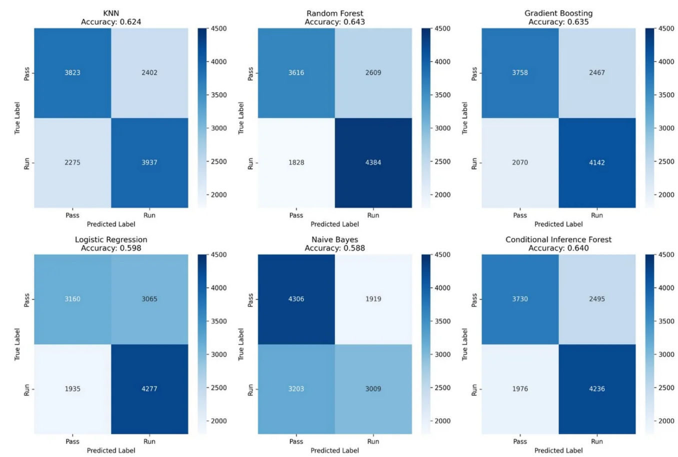

Fig. 6. Confusion matrices for all six machine learning models.

plays incorrect. Overall, this model is balanced in correct and incorrect way without over correcting for certain play types. This can be shown in its heatmap where it is the most balanced, showing it did not over predict one type of play.

Random Forest performs the strongest overall, with 3,616 correctly identified pass plays and 4,384 correctly identified run plays. It incorrectly predicts 2,609 pass plays and 1,828 run plays, giving it the lowest error count among all models. Unlike K-Nearest Neighbor, Random Forest shows a bias towards predicting run plays more accurately while pass plays are noisier, with it getting slightly more pass plays incorrect than K-Nearest Neighbor.

Gradient Boosting shows strong performance, correctly predicting 3,758 pass plays and 4,142 run plays. It incorrectly predicts 2,467 pass plays and 2,070 run plays, placing it close behind Random Forest and Conditional Inference Forest. Gradient Boosting demonstrates a similar preference toward accurately identifying run plays, while pass predictions are slightly more error prone. It doesn't accurately get run plays as much as accurate as random forest, but it performs slightly better in passing play scenarios. In its heatmap, this appears as a darker run-correct cell and a lighter pass-correct cell, showing that while it performs well overall, Gradient Boosting does not fully eliminate noise in pass-heavy scenarios.

Logistic Regression performs noticeably weaker than the tree-based methods, correctly identifying 3,160 pass plays and 4,277 run plays. It incorrectly predicts 3,065 pass plays and 1,935 run plays, showing difficulty accurately capturing pass-play patterns. Logistic Regression tends to lean toward predicting runs more reliably, while its pass prediction is much worse. This can be seen in the heatmap through a lighter shading in the pass-correct cell, indicating a reduced ability to predict passing situations compared to other models.

Naïve Bayes shows the weakest performance overall, with 4,306 correctly predicted pass plays but only 3,009 correctly predicted run plays. It incorrectly predicts 1,919 pass plays and 3,203 run plays, revealing a clear bias toward predicting pass plays. This imbalance is clearly visible in its heatmap, where the pass-correct region is significantly darker than the run-correct region. This is the only model in which more accurately predicts pass plays than run plays. Overall, this model very heavily relies on assuming the play will be a pass, likely due to its issue with assuming feature independence.

Conditional Inference Forest performs at a high level, like the other forest models. It correctly identifies 3,730 pass plays and 4,236 run plays. It incorrectly predicts 2,495 pass plays and 1,976 run plays, giving it one of the most balanced error profiles among the top performers. Similar to Random Forest, it shows stronger accuracy on run plays, though it handles pass plays slightly better than some other models. Its heatmap reflects a clear, consistent diagonal pattern, indicating stable and reliable predictions across both classes without major overprediction of one play type though not as well as K-Nearest Neighbors balance.

Overall, this helps show where each model is better and worse at predicting plays. The best overall was Random Forest, which so happened to also get the most run plays correctly. In contrast Nieve bayes is the worst overall, but it actually predicts pass plays the best, this however is mainly due to its over reliance on feature independence, causing it to over predict pass plays. Most models performed relatively similar to one another however, with run being the easier play call to predict for most models. However, different models perform different in different in-game scenarios.

F. Situational Performance

Across almost all the models, prediction accuracy increases substantially in various more constrained scenarios. This shows scenarios where play-calling tendencies are more predictable. Accuracy increases substantially in constrained scenarios such as 3rd and 4th down, short-yardage situations, goal-to-go, and red-zone plays. These patterns reflect clearer offensive strategy when play-calling becomes more predictable, this is also observed in sequence-based football models [5]. For example, 4th and 1 becomes obvious it will likely be a run play, while if a team is desperate at the end of the game and it is 4th and 8, it will be obvious the team will pass. Similarly, goal-to-go, red-zone, and 4th quarter scenarios consistently show higher accuracy, reflecting clearer offensive strategy when the game situation gets more constrained, and play-calling becomes more obvious. Fig. 7 presents the situational performance heatmap across all models and game situations.

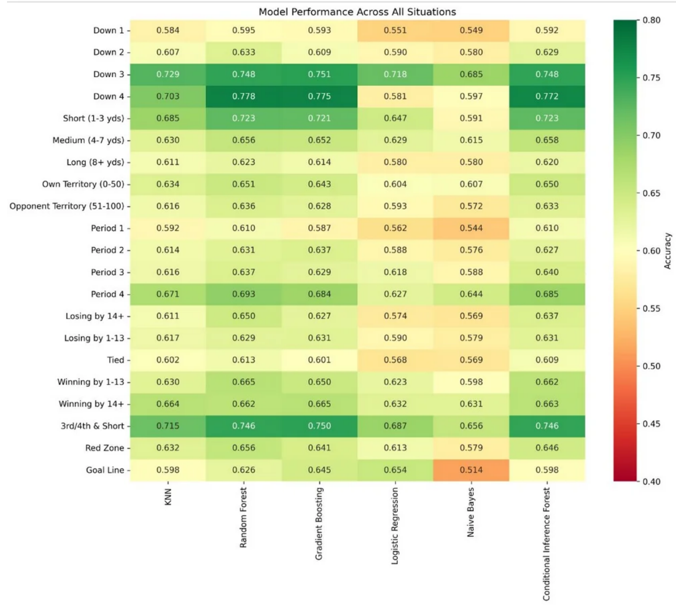

Fig. 7. Model performance heatmap across all situational contexts.

V. Analysis

The performance across all models reveals several consistent patterns between the different models and strongly suggests a hierarchy between each of the models, with the order being Random Forest, Conditional Inference Forest, Gradient Boosting, K-Nearest Neighbor, Logistic Regression, and Naïve Bayes. In every overall performance metric showed this to be the case. However, there was some variation when looking at the model's performance to access specific scenarios, and prediction accuracy on confusion matrices showed different strengths and weaknesses of each model, which we will discuss.

In overall performance, when measuring accuracy, recall, precision, and F1-score, each model followed a hierarchy, with little nuance. It was pretty clear each model was in its specific spot in the hierarchy with the order being Random Forest, Conditional Inference Forest, Gradient Boosting, K-Nearest Neighbor, Logistic Regression, and Naïve Bayes. The largest takeaway from this data was the overall advantage tree-based models had on this data. All the top three models are tree-based with those being random forest, conditional inference forest, and gradient boosting. This strongly suggests that overall performance on categorical noisy college football play data is particularly well suited for decision tree models. While the other models are less accurate, such as k-nearest neighbor, logistic regression, and naive bayes.

The clear hierarchy in model performance can be attributed to how well each model captures non-linear relationships among features. Tree-based models excel at modeling complex interactions between situational factors, which has been noted in prior work on machine learning applications in football, and their performance can be systematically evaluated using metrics like accuracy, precision, recall, and F1-score [4]. As seen by how many different metrics go into a play call decision, college football play calling is highly dependent on complex relationships between multiple different factors. For instance, if a team is losing it is 3rd and 2 at the opponent's 41 with 30 seconds remaining, the logical play call would be to pass the ball. While if a team is losing it is 3rd and 2 at the opponents 41 with 5 minutes remaining, the logical play call would be to run the ball. As you can imagine, there are multiple things in a coach's head when calling a play that affects the call. Tree-based models naturally capture the interactions between all of these complex relationships between down, distance, field position, time remaining, score differential, and more. This allows these models to learn patterns such as short yardage, red zone, late game, and other tendencies that happen in specific scenarios. On the other hand, logistic regression and naive bayes assume linearity, also known as feature independence, which isn't the case in college football play calling where different "features" interact and influence the overall outcome of the play call. The situational heatmap provides a comprehensive view of how each model behaves under a large range of football situations. This shows patterns that are not visible through overall accuracy alone. Through showing different football scenarios, a deeper insight into how the models behave in complex scenarios is shown. A clear trend across all models is that prediction of accuracy increases as the offensive decision space narrows. For example, the highest accuracy values appear consistently on 3rd and 4th down, especially in short-yardage situations which is between 1–3 yards. This makes sense, as your distance, and different parameters are minimized; it makes playcalling a little more predictable and obvious for both the coach and our models. For example, playcalling becomes heavily biased toward running the ball once there are fewer attempts, and the yardage is shorter. As has been previously talked about, tree-based models excel in this type of data, and this is further shown. Tree-based models, such as the Random Forest, Conditional Inference Forest, and Gradient Boosting, reach their best performance here, with accuracy values exceeding 0.75 for each. This minimizing of different constraints shows a strong ability to capture these predictable tendencies since 3rd and 4th down in short yardage situations. Similarly, accuracy tends to increase in constrained scoring situations such as the red zone and goal line, where teams rely more on high-percentage plays, and the available options are inherently limited. This can again be because as the field shortens it becomes harder to throw the ball, and especially at the goal-line, running the ball on short yardage becomes the easy choice to make. In contrast, accuracy is lower in more flexible situations such as 1st down, medium-distance plays which is between 4–7 yards, and tied games, where offenses have a wider range of viable strategies and exhibit more variability in play selection. In scenarios where there are more active parameters that can be considered, this causes the play calls to become more inconsistent and varied. When a team is down for example it makes more sense to pass the ball, but if the teams are tied the offense has the ability to choose whatever they desire. The models also show shifts in performance depending on game flow. For example, when a team is winning by a large margin, accuracy rises because teams often shift toward run-heavy offenses as clock-management strategies become more desired, which is correlated to running the ball. On the other hand, when losing, accuracy tends to fall as teams adopt unpredictable passing-heavy approaches as if the ball is dropped the play clock stops and it allows for longer yardage plays. The quarter-by-quarter breakdown mirrors this pattern, with moderate performance early in the game and higher accuracy in the fourth quarter, where situational game scenarios and predictable play-calling tendencies become more pronounced like the running the ball or passing the ball based off score differential. Across every situation, the tree-based models reliably outperform the others, while Naïve Bayes and Logistic Regression struggle due to their issues discussed during this paper. Overall, the heatmap shows that run-pass predictability is strongly tied to situational constraints, and that model performance is highly tied to these different scenarios.

With all of the different performance metrics, the combined results show a clear pattern and obvious benefit of some models. The consistent hierarchy observed across accuracy, precision, recall, and F1-score, by the patterns in the confusion matrices and situational breakdown all show that tree-based methods are the most accurate and the most adaptable to the contextual complexity of college football playcalling. This is due to their ability to incorporate non-linear feature interactions, which in turn allows them to generalize effectively across all game scenarios. These outperform simpler models. Logistic Regression and Naïve Bayes are less accurate because the underlying structure of their algorithms conflict with the noisy data in college football play calls. Across all analyses, a clear pattern is shown emerges as model performance improves as the decision space becomes more restricted while when there is a larger decision space the models perform more inaccurately. This indicates that while machine learning can meaningfully capture structured tendencies in football data, it is bounded by the uncertainty of real-time gameplay decisions. Overall, this shows both the potential and the limitations of the models shown in this, and the data given to them.

VI. Conclusion and Future Work

This paper proposed machine learning frameworks to classify college football plays between a binary run pass application, using play data from ESPN's API for the college football 2024 season. This framework consists of two data scrapers. One for getting all game IDs from ESPN, and then a second utilizing the list of game IDs to get play by play data for all Power 4 games during the 2024 College football season. After data processing, multiple machine learning techniques were utilized to compare and identify the best model based off certain statistics such as overall accuracy.

The primary contribution of this work is to establish a framework for predictive analytics. This framework generates a large-scale dataset that can be easily used in multiple different frameworks. This paper also utilizes multiple including Random Forest, K-Nearest Neighbor, Naive Bayes, Logistic Regression, Gradient Boosting, and Conditional Inference Trees different machine learning techniques, which showcase how different models can have their upsides, as well as their downsides. These models were then tested on a variety of different metrics including key statistics like accuracy, precision, recall, and F1-Score. Other statistics such as prediction accuracy, and true positive/true negative results are key to this paper along with specific feature analysis to determine model performance across different situations. This comprehensive testing suite developed in this paper is key to showing refinement or potential future work as this gives a baseline and way to evaluate potential future work.

Future work on this topic would be expanding the scale and complexity. This paper and data focused only on Power 4 teams; this could be expanded to the group of 5, and potentially down to other lower tiers such as the FCS, Division 2, and Division 3. While likely yielding different results due to the differences in how the game is played, and some rule differences, this could give a larger sample on what is effective. Another major area of improvement is adding complexity. Instead of a binary pass run method used in this paper, adding extra complexity where it considers if it's a short, medium, or long pass or if the run play is inside the tackles or outside the tackles. Another way to add complexity is through advanced statistics not provided by the ESPN API. This added data would help contextualize other data provided as well as give more data to hopefully boost accuracy and effectiveness. In future work, incorporating continuous-time player-tracking data could provide richer context for play-call prediction by capturing spatial-temporal dynamics, individual player movements, and interaction patterns across the field. Approaches like those demonstrated by Yurko [7] show that such detailed, fine-grained information can significantly enhance predictive accuracy and enable models to account for in-play variability that is not captured by traditional situational features. For example, this model does not include any information about the defensive rating that the team is facing, which will decrease the accuracy of the model. This could also include individual players that are playing within the game. Injuries and player personnel on the field at a given time are a part of football, so week-to-week injuries and play-by-play player personnel will impact how a team will decide to run offense, which is not modeled in this. Giving each model context on who is playing, and who is injured would more strongly help model the offense. If your starting quarterback is injured, there will likely be more runs, which this model does not model at all. Extra models could also be trained on this information to again find where there are strong points for each model, like found in this paper's results. In this paper's findings, there was a large amount of data that points to tree-based models having a stronger overall accuracy, which could then be used to add extra fine-tuning to the already given models in this paper or explore other tree-based models. Other very commonly tried models such as neural networks could also be used for this dataset to see how it performs compared to the other models used in this paper, however the main purpose of this paper was to explore a variety of models and not narrow it down to just one model, or one type. Overall, added complexity and depth would be strong areas to improve or extend this paper.

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